products. According to the analysts there would be a steady increase in the adoption in the coming year. Following this trend, some business software companies have developed AI capabilities in-house, but many others are acquiring capabilities through mergers and acquisitions, this trend is expected to continue in 2016. Strong support from venture capital investors is also helping to further commercialise this market. Since 2011, US-based startups that develop or apply cognitive technologies to enterprise applications have raised nearly US$ 2.5 billion, suggesting that the biggest near-term opportunity for cognitive technologies is in using them to enhance business practices.

Implementation challenges

The capabilities around cognitive computing and the market's expectation from them are driving the need for a new class of supercomputer systems, which can enable the synergy of advanced analytics and Big Data technology. However, there are challenges that we need to be aware of. Cognitive computing technologies are quite complex, inherently. The complexities imply steep learning curves and, hence, increased turnaround times. Building capabilities around these systems requires considerable amount of time and effort.

There are many solutions involving these technologies from various leading vendors. Many of the platforms and the frameworks, however, are extremely expensive and call for heavy infrastructure. Implementation of cognitive automation calls for huge amounts of investment.

Open source tool sets—a probable solution

Till recently, cognitive computing was confined more to the academic world. That could be the reason why a huge number of open source tools and libraries have evolved around cognitive computing and related technologies. Today, a wide range of solutions, along with the huge knowledge and code base, are available in the open source domain.

This rich repository can enable enterprises to learn and experiment with these technologies, thus increasing their reach. These solutions will allow enterprises to create a quick prototype around cognitive computing. This way, enterprises can check the viability of the underlying idea and get quick feedback about it.

Some of the popular open source solutions in the area of cognitive computing are:

- **R:** R is a language and environment for statistical computing and graphics. It provides a wide variety of statistical (linear and non-linear regression, classical statistical tests, time-series analysis, classification, clustering, etc) and graphical techniques. It is highly extensible. The R language is often the vehicle of choice for research in statistical methodology, and R provides an open source route to participation in that activity.
- **Python:** While R is specifically created for statistical analysis, Python also has a rich set of machine learning implementations. It is widely used among the scientific community. Being an interpreter, high-level programming language, Python is a good fit for machine learning implementation, as quite often, this calls for an agile and iterative approach.
- **Apache Mahout:** This provides an environment for quickly creating scalable machine learning applications.
- **H2O:** H2O is for data scientists and application developers who need fast, in-memory, scalable machine learning for smarter applications. H2O is an open source parallel processing engine for machine learning.
- **RapidMiner:** RapidMiner is a platform that provides an end-to-end development environment for machine learning. Through a wizard-driven approach, RapidMiner allows the user to rapidly build the predictive analytics model.

Points to ponder

To check the viability of a business case, it would be a good idea to first take baby steps rather than taking a huge plunge. Using open source solutions, enterprises can avoid upfront investments and check the viability of various cognitive technologies.

To check the viability, first we need to identify the right use case. We should have thorough understanding of the business problem it attempts to solve. This has to be followed by right selection of the technology solution suitable for implementation.

As seen in the previous section, a wide range of cognitive computing solutions are available as open source. Each solution has evolved to cater to a specific set of business problems. Each one of them has its own target user group. So to get the desired result, we need to first have a clear understanding of the problem at hand. Based on that understanding, we need to choose the right tool set to best solve that type of business problem. With a quick prototype on selected small set of use cases implemented with right choice of open source solutions would enable the enterprise to check the viability of the cognitive automation initiative. END

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